

Saumya Mehta

AI Engineer

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Senior AI Engineer with 6+ years designing and shipping production LLM, RAG, and recommendation systems across edtech, banking, and healthcare. Presented at ICICLE-25 on large-scale experimentation → 1.2 × faster iteration for educators. Proven ability to operationalise AI solutions, communicate complex technical ideas to non-experts, and lead cross-functional initiatives.

WORK EXPERIENCE

Bank Of America <i>Data Scientist, Machine Learning & AI</i>	Sept 2025 - Present Plano, TX
- Architected a multi-task learning framework to classify 21 document types across 4 sources, using PCgrad and homoskedastic uncertainty weighting to resolve conflicting optimization paths. This raised straight-through processing to 49% (at 95% confidence) and increased document accuracy from 67% to 76%. - Developed a multi-agent system using group deliberation methods for LLM trustworthiness, employing an ensemble of models (LLaMA 3.1 70B, GPT-OSS) to calibrate confidence. Achieved significant performance gains: 35% reduction in Expected Calibration Error (ECE) and 20% improvement in the Brier score.	
Aseesa Inc <i>AI Research Engineer</i>	Sept 2025 - Present Plano, TX
Architected a scalable RAG pipeline to construct a comprehensive disease knowledge base from over 5k scientific articles sourced from PubMed and Crossref, establishing a high-fidelity corpus for downstream biomedical research. Built and tuned a low-latency vector search stack on Qdrant, indexing over 10k documents with 50-dim embeddings and rich payload filters, sustaining 5,000+ vectors per second during ingestion and delivering p95 query latency under 40 ms while improving precision@10 by 20% over a BM25 baseline.	
Carnegie Learning <i>Senior Data Scientist, Machine Learning & AI</i>	June 2023 – July 2025 Pittsburgh, PA
- Led RCTs and A/B tests across 10K+ students in partner school districts, delivering insights that drove core MATHstream feature improvements and informed district-level product decisions. - Designed and deployed scalable logging infrastructure to capture and standardize 500K+ student events daily in real time, enabling robust analytics and cross-team research integration. - Conducted quasi-experimental analysis to assess MATHstream's impact on standardized test scores, revealing 4–7% gains in standardised test outcomes resulting in 15% upsell conversion in pilot districts (est 1.7B revenue) - Productionized real time recommendation algorithms as Python FastAPI microservices with Pydantic validation and pytest suites; reducing p95 latency by 57% using batching, caching, and lightweight quantization Built scalable vector indexing and semantic search systems leveraging Pinecone and OpenAI embeddings, optimizing document chunking, similarity retrieval, and response latency for downstream LLM tasks. - Built Python NLP pipelines for NER, document classification, and summarization to index curriculum and parse student responses, powering recommendations and adaptive hint generation; HR@5 +12%, NDCG@10 +9% - Built PySpark + Airflow pipelines for recency/mastery/difficulty features; improved data freshness from hours to minutes, lifting rec quality on new content. Designed and deployed Retrieval-Augmented Generation (RAG) pipelines using LangChain, OpenAI embeddings, and PGVector to enable contextual response generation from proprietary document datasets.	
Pearson(Savvas) <i>Software Development Engineer Intern</i>	May 2022 – Aug 2022 Boston, MA
- Developed Curriculum Recommendation algorithms for the SuccessMaker engine using Bayesian Item Response Theory resulting in a 40% improvement in student test scores - Streamlined the data analytics pipeline for the Learning Analytics team by creating external views in Amazon Redshift DB and automated query scheduling on AWS for ETL tasks	
Playpower Labs <i>Data Scientist</i>	Jun 2018 – Aug 2021 Remote
- Led the research and development of an AI-powered paper-based formative learning product aimed at creating equitable assessments for students across India and piloted across 50+ schools - Built an OCR/Hand written text extraction pipeline with de-skew/denoise and region segmentation to extract free text responses from paper assessments; achieved 3–5% CER on clean scans and 6–8% CER on classroom scans using lexicon-based post-correction. - Architected and deployed end-to-end machine learning pipelines on Snowflake, handling up to 1 million records per second and delivering a 25% improvement in model performance and a 50% reduction in data processing time - Enhanced real-time data processing capabilities by utilising Spark Streaming, R, and Kafka, delivering a 50% improvement in processing speed and scalability for a distributed computing setup handling over 100K records per second of streaming data - Implemented and optimized ETL pipelines using Apache Airflow and AWS Glue, delivering a 30% reduction in ETL runtime and improving data quality and reliability - Implemented custom OpenCV kernels with native functions to deliver sub-second page detection, region extraction, and answer extraction for both free-text and MCQ responses, processing 10–20 page workbooks in under 30 seconds with around 98% end-to-end grading accuracy - Researched and applied model quantization techniques for character-recognition models, reducing model footprint enough to run reliably on ultra-low-end Android phones in India (sub \$50 devices).	

PUBLICATIONS

Gastroenterology in the age of AI: Bridging technology and clinical practice [doi: 10.3748/wjg.v31.i36.110549]	Sept 2025
<i>World Journal of Gastroenterology</i>	
MATHstream and UpGrade: Using Rapid, Large-Scale Experimentation for Data-Driven Improvements [Paper]	June 2025

<i>International Consortium for Innovation and Collaboration in Learning Engineering (ICICLE)</i>	
Simvastatin + Metformin Treatment Targets Growth and Fibroinflammatory Responses in PaSc [Poster]	Dec 2023
<i>American Pancreatic Association</i>	
Advanced Chemical Transport Modeling in Dynamic Multicellular Contexts Using CompuCell3D [Manuscript]	May 2023
<i>BioPhysical Journal</i>	

TECHNICAL SKILLS

Languages & Frameworks: Python, R, PostgreSQL, Redis,, C++, Streamlit, Django, Pytorch, Tensorflow, SageMaker, Ollama

AI & Machine Learning: Prompt Engineering, Reinforcement Learning, Quasi Experimental Analysis, RAG, A/B Testing, Causal Inference, Bayesian Inference, Model Interpretability, LangChain, LangGraph, a2a, DeepEval, FAISS, Azure AI, Huggingface

Data Pipelines: Spark, Kafka, Snowflake, Docker, Kubernetes, Jenkins, AWS, Databricks

EDUCATION

Indiana University	Bloomington, IN, USA
<i>Masters of Science in Data Science CGPA: 3.93</i>	Aug 2021 – May 2023
Nirma University	Ahmedabad, GJ, IN
<i>Bachelors of Science in Computer Science CGPA: 7.8/10</i>	Aug 2014 – June 2018

PROJECTS

- Improving Image Captions with Depth Maps** [\[Github\]](#)
 - Improved image captioning using CNN and RNN models and incorporating depth maps, for accurate and informative captions.
- Abusive Language Detection in Social Media using Natural Language Processing** [\[Github\]](#)
 - Developed a multi headed model capable of detecting abusive language and threats like hate speech, obscenity, targeted threats using LSTMs and glove-emoji embeddings
- Sliding and Picking objects with Robot Arm using Reinforcement Learning** [\[Github\]](#)
 - Developed a Q-learning-based control algorithm and implemented a hindsight replay buffer for object manipulation using a robot arm, resulting in a **20-25%** increase in success rates on pushing and sliding tasks.